Lab -4: **Experiment No: 4** **Dated:**

AIM: To develop a blockchain-powered web application using Solidity programming language on Remix IDE and embedding Metamask Wallet

***Course Outcomes (CO)*:**

Task to be performed: **[Follow the GitHub repository for the experiment]**

1. Set Up MetaMask:
   a. Install MetaMask
   b. Create or Import an Account:
   c. Fund Your Wallet : Sepolia Testnet (0.5 ETH per day) / RSK Testnet (0.05 RBTC per day)
2. Connect the Sepolia Testnet / RSK Testnet to Remix IDE
3. Create a Simple Solidity Smart Contract **based on the MiniPoject chosen**
4. Compile and Deploy the Smart Contract.
5. Check the transaction details on the RSK Explorer
6. Interact with the smart contract

Instructions : (Prepare for viva for the following topics)

1. What is a Metamask?
2. What is a test net?
3. List the steps to connect a Metamask with a Remix IDE for performing transactions.

Outcome :

1. Understood the steps for embedding the Metamask wallet with Remix IDE and perform transactions
2. Successfully performed the transactions on the Remix IDE via the account from Metamask Wallet
3. Prepare a document with the Aim, Tasks performed, Program, Output, and Conclusion.
4. Submit the hard copy **by the 5th week of September**
   (As per the instructions, submit a hard copy of the same)

Aim

To develop and deploy a blockchain-powered web application using Solidity on Remix IDE, integrate it with the MetaMask wallet, and perform smart contract transactions on the Sepolia test network.

Theory

A smart contract is a self-executing program stored on a blockchain. It contains predefined rules and executes automatically when its functions are called. Smart contracts are commonly written using Solidity for Ethereum-compatible blockchain networks.

1. Solidity

Solidity is a programming language used to create smart contracts for Ethereum and other EVM-compatible blockchains.

In this experiment, a Solidity smart contract called `ProductOwnership` was created to:

- Store a product name.
- Store the owner's blockchain address.
- Transfer ownership to another address.
- Retrieve the product name and current owner.

2. Remix IDE

Remix IDE is a browser-based development environment used to write, compile, deploy, and interact with Solidity smart contracts.

The basic workflow is:

Write Contract → Compile → Connect Wallet → Deploy → Interact

3. MetaMask

MetaMask is a cryptocurrency wallet and browser extension that allows users to interact with blockchain networks and decentralized applications.

In this experiment, MetaMask was used to:

- Connect the user's blockchain account to Remix.
- Select the Sepolia test network.
- Approve smart contract deployment.
- Pay transaction fees using SepoliaETH.

4. Testnet

A testnet is a blockchain network used for testing applications and smart contracts without using real cryptocurrency.

Sepolia was used in this experiment. The ETH available on Sepolia is test cryptocurrency and is used to pay gas fees during testing.

5. Gas and Transactions

When a smart contract is deployed or a state-changing function is executed, computational resources are required. The fee paid for these resources is called **gas**.

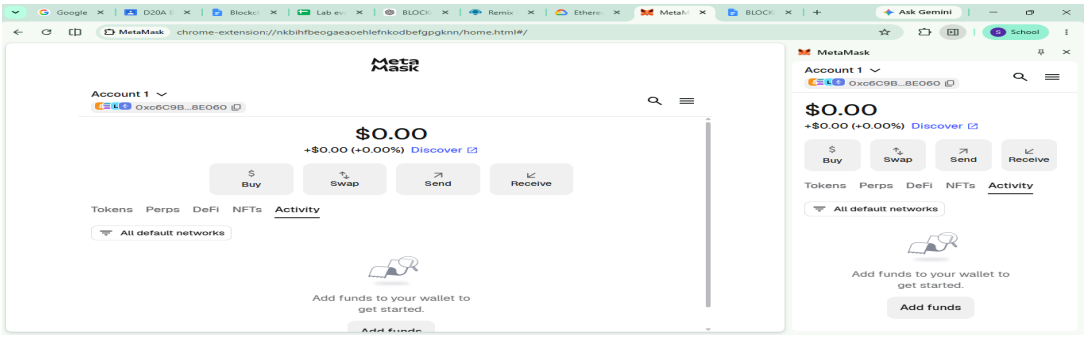
For example, deploying the `ProductOwnership` contract required a small amount of SepoliaETH as a network fee.

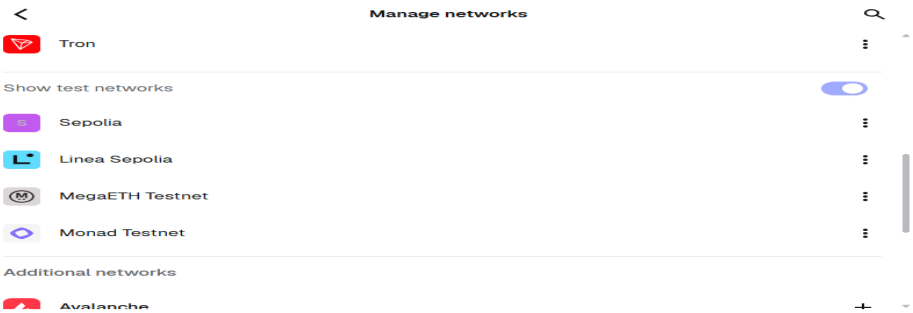
6. Smart Contract Used

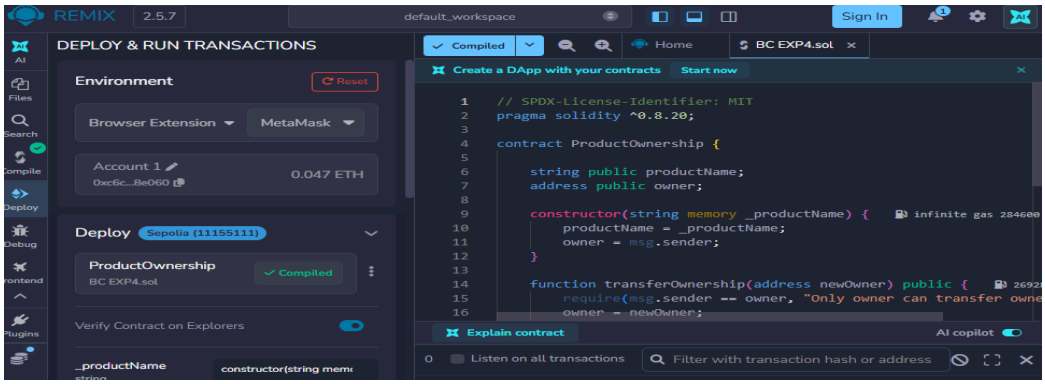
The contract contains:

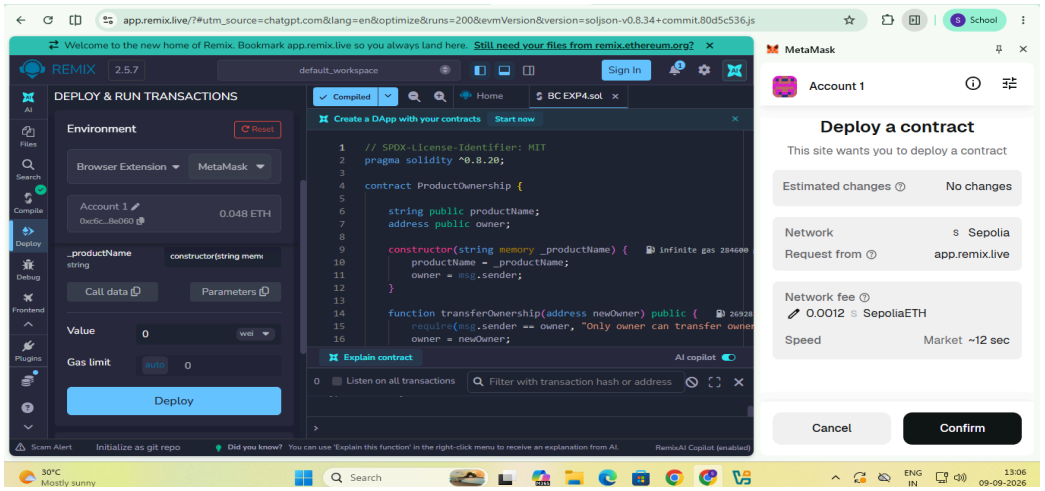
- `productName`: Stores the name of the product.
- `owner`: Stores the current owner's wallet address.
- `constructor()`: Sets the product name and assigns the deploying account as the initial owner.
- `transferOwnership()`: Transfers ownership to another blockchain address.
- `getProductDetails()`: Returns the product name and current owner.

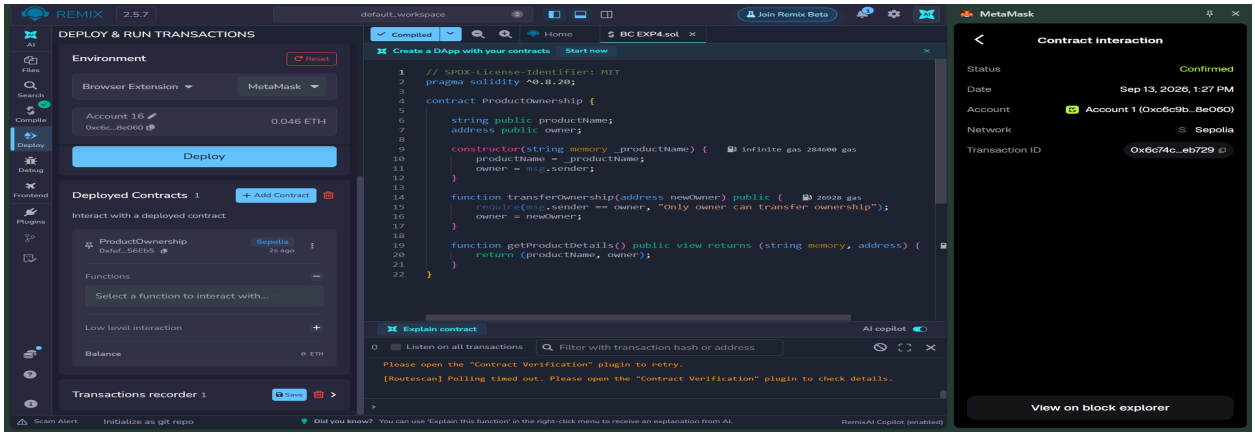
The contract was compiled using **Solidity 0.8.20**, deployed through Remix, and approved through MetaMask on Sepolia

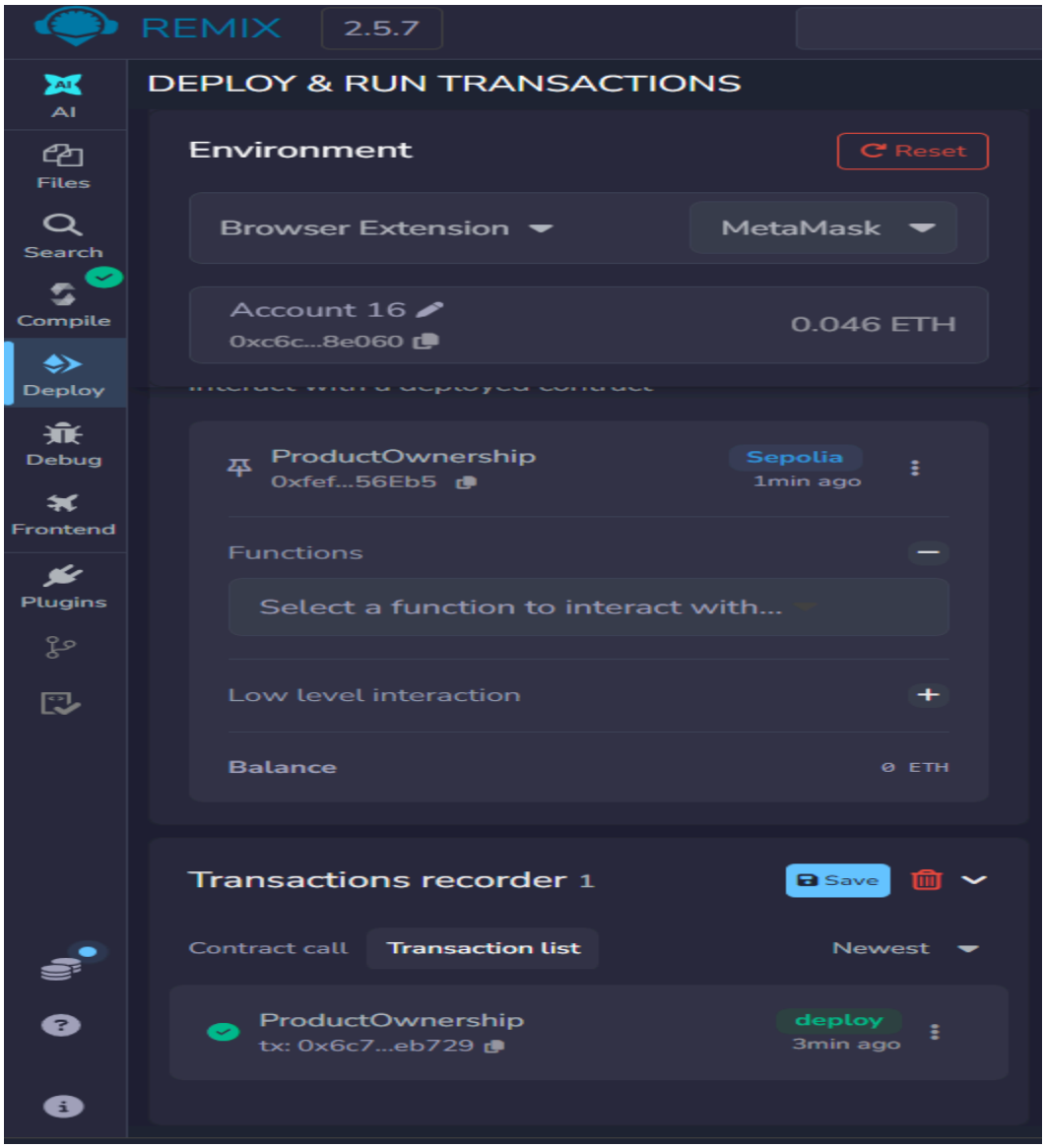



Manage networks
Tron
Show test networks
Sepolia
Linea Sepolia
MegaETH Testnet
Monad Testnet
Additional networks
Avalanche


REMIX 2.5.7
default_workspace
Sign In
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
MetaMask
Account 1
0xc6c...8e060
0.047 ETH
Deploy
Sepolia (11155111)
ProductOwnership
BC EXP4.sol
Compiled
Verify Contract on Explorers
_productName
constructor(string memo
Home
BC EXP4.sol
Create a DApp with your contracts
Start now
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract ProductOwnership {
string public productName;
address public owner;
constructor(string memory _productName) {
infinite gas 284600
productName = _productName;
owner = msg.sender;
}
function transferOwnership(address newOwner) public {
require(msg.sender == owner, "Only owner can transfer owne
owner = newOwner;
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or address


app.remix.live/?#utm_source=chatgpt.com&lang=en&optimize&runs=200&evmVersion&version=soljson-v0.8.34+commit.80d5c536.js
Welcome to the new home of Remix. Bookmark app.remix.live so you always land here. Still need your files from remix.ethereum.org?
REMIX 2.5.7
default_workspace
Sign In
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
MetaMask
Account 1
0xc6c...8e060
0.048 ETH
_productName
string
constructor(string memo
Call data
Parameters
Value
0
wei
Gas limit
auto
0
Deploy
Compiled
Home
BC EXP4.sol
Create a DApp with your contracts
Start now
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract ProductOwnership {
string public productName;
address public owner;
constructor(string memory _productName) {
infinite gas 284600
productName = _productName;
owner = msg.sender;
}
function transferOwnership(address newOwner) public {
require(msg.sender == owner, "Only owner can transfer owne
owner = newOwner;
Explain contract
AI copilot
Listen on all transactions
Filter with transaction hash or address
Scam Alert
Initialize as git repo
Did you know?
You can use 'Explain this function' in the right-click menu to receive an explanation from AI.
RemixAI Copilot (enabled)
MetaMask
Account 1
Deploy a contract
This site wants you to deploy a contract
Estimated changes
No changes
Network
Sepolia
Request from
app.remix.live
Network fee
0.0012 SepoliaETH
Speed
Market ~12 sec
Cancel
Confirm
30°C
Mostly sunny
Search
ENG
IN
13:06
09-09-2026


REMIX 2.5.7
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
MetaMask
Account 16
0xc6c...8e060
0.046 ETH
Deploy
Deployed Contracts 1
Add Contract
Interact with a deployed contract
ProductOwnership
0xfef...56Eb5
Sepolia
2s ago
Functions
Select a function to interact with...
Low level interaction
Balance
0 ETH
Transactions recorder 1
Save
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.20;
contract ProductOwnership {
string public productName;
address public owner;
constructor(string memory _productName) {
productName = _productName;
owner = msg.sender;
}
function transferOwnership(address newOwner) public {
require(msg.sender == owner, "Only owner can transfer ownership");
owner = newOwner;
}
function getProductDetails() public view returns (string memory, address) {
return (productName, owner);
}
}
Please open the "Contract Verification" plugin to retry.
[Routescan] Polling timed out. Please open the "Contract Verification" plugin to check details.
MetaMask
Contract interaction
Status
Confirmed
Date
Sep 13, 2026, 1:27 PM
Account
Account 1 (0xc6c9b...8e060)
Network
Sepolia
Transaction ID
0x6c74c...eb729
View on block explorer


REMIX
2.5.7
AI
Files
Search
Compile
Deploy
Debug
Frontend
Plugins
DEPLOY & RUN TRANSACTIONS
Environment
Reset
Browser Extension
MetaMask
Account 16
0xc6c...8e060
0.046 ETH
ProductOwnership
0xfef...56Eb5
Sepolia
1min ago
Functions
Select a function to interact with...
Low level interaction
Balance
0 ETH
Transactions recorder 1
Save
Contract call
Transaction list
Newest
ProductOwnership
tx: 0x6c7...eb729
deploy
3min ago

Conclusion

The experiment was successfully completed by developing a Solidity-based smart contract using Remix IDE and connecting it with the MetaMask wallet. The `ProductOwnership` contract was compiled and deployed on the Sepolia test network. MetaMask was used to approve the deployment and pay the required testnet gas fee. The deployed contract could then be interacted with to retrieve product information and manage ownership.